1 Introduction

1.0 Document Purpose

This Software Requirements Specification (SRS) describes the functional and non-functional requirements for the Centralized Web-Based System for Stress Induction Methods.

The purpose of this document is to provide a detailed description of the system's functionality, performance requirements, and operational constraints. The system is designed to assist researchers in conducting stress-induction experiments by providing a centralized platform for managing participants, administering stress-induction methods, recording responses, and generating experiment data.

This document serves as a reference for developers, researchers, project supervisors, testers, and other stakeholders involved in the development and implementation of the system. It ensures that all parties have a clear understanding of the system's objectives, features, and requirements.
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1.1 Product Scope

The Centralized Web-Based System for Stress Induction Methods is a web-based platform developed to support researchers in conducting stress-induction experiments in a structured and efficient manner.

The system centralizes different stress-induction methods into a single platform where researchers can manage participant information, conduct experimental sessions, and collect research data. By digitizing these processes, the system helps reduce manual errors, improve data organization, and enhance the efficiency of research activities.

The platform allows researchers to create participant profiles, initiate stress-induction tests, monitor experiment progress, and store experiment results in a centralized database. This centralized approach ensures better accessibility, consistency, and management of research data.

1.2 Intended Audience and Document Overview

This SRS document is intended for stakeholders who are involved in the development and use of the Centralized Web-Based System for Stress Induction Methods.

- **Developers**: Developers will use this document as a guideline for designing, implementing, and maintaining the system according to the defined requirements.
- **Researchers**: Researchers will use the system to manage participants, conduct stress-induction experiments, and collect research data.
- **Project Supervisors**: Project supervisors will review this document to ensure that the system meets the research objectives and project requirements.
- **Testers**: Testers will use this document to verify whether the system meets the specified functional and non-functional requirements.
- **Participants**: Participants are individuals involved in the stress-induction experiments who interact with the system during experimental sessions.

1.3 Definitions, Acronyms and Abbreviations

Terms	Definition
SRS	Software Requirements Specification.
Researcher	A user responsible for conducting experiments and managing research data
Participant	An individual who participates in the stress-induction experiment
Stress Induction Method	A controlled experimental technique used to induce stress for research purposes
Experiment Session	A session where participants complete stress-induction tasks
System	Centralized Web-Based System for Stress Induction Methods.

1.4 Document Conventions

This document follows the IEEE Software Requirements Specification (SRS) standard format. The document is organized into sections that describe the system overview, functional requirements, non-functional requirements, and system features.

Consistent formatting and terminology are used throughout the document to ensure clarity and readability for all stakeholders involved in the project.

1.5 References

The development of the Centralized Web-Based System for Stress Induction Methods is based on the requirements of the Final Year Project and research materials related to stress-induction methodologies.

2 Overall Description

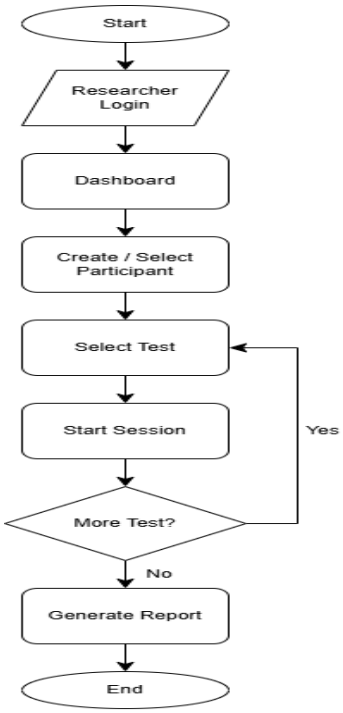
2.0 Product Perspective

The Centralized Web-Based System for Stress Induction Methods is a centralized system developed to assist researchers in conducting stress-induction experiments efficiently and effectively. The system aims to improve the current methods of conducting the test which the current method uses manual stress test and manual recording.

By centralizing the stress-induction methods in one single platform, the system allows researchers to manage the participant, conduct different type of test in one platform which can reduce the risk of data loss, inconsistency, and human error.

They system consists of the following major components:

- User Management Module: Handles authentication and role-based access control for system users, including administrators, and researchers.
- Participant Management Module: Stores participant information such as participant ID, participant information, and session records.
- Experiment Session Module: Guide participants through the experiment process by presenting stimuli, recording responses, measuring reaction times, and monitoring sessions progress.
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- Data Storage and Processing Module: Stores experiment data securely and performs automatic calculations such as reaction time averages, error rates, and stress scores.
- Reporting and Export Module: Generates experiment information and allows researchers to export raw data and reports in formats such as CSV for analysis and PDF for summary report.



2.1 Product Functionality

The Centralized Web-Based System for Stress Induction Methods offers the following major functions:

1. User Management:

    - Allow researchers and administrators to securely log into the system.
    - Manage user roles such as administrator and researcher.
    - Control access permissions to ensure only authorized users can configure experiments or export data.

2. Participant Management:

    - Create and store participant profiles using pseudonymous participant IDs.

3. Stress Test Management:

    - Guide researchers through the experiment process step-by-step.
    - Present stress-induction stimuli to participants during the session.
    - Monitor experiment progress in real time.
    - Allow participants to repeat experiment sessions while storing each session separately.

4. Reporting:

    - Generate automatic experiment summary reports.
    - Provide raw data.
    - Allow export in formats such as CSV and PDF.

2.2 Users and Characteristics

The Centralized Web-Based System for Stress Induction Methods is designed to be functioned to multiple users with different roles and responsibilities. The characteristics of these users are detailed below:

1. Administrator

    - Characteristics: Responsible for managing system configuration, user accounts, and system settings. The administrator ensures that the platform operates securely and efficiently.
    - Frequency of Use: Moderate
    - Privilege Levels: Full administrative access, including managing researcher accounts, configuring system settings, monitoring system activity, and maintaining data security.
    - Technical Expertise: Advanced knowledge of system administration and database management.

2. Inspector

    - Characteristics: Responsible for conducting stress-induction experiments using the system. Researchers configure experiment templates, manage participant information, initiate experiment sessions, and analyze experiment data.
    - Frequency of Use: High

- Privilege Levels: Authorized to create and manage experiment templates, manage participant records, conduct experiment sessions, view experiment results, and export research data.
- Technical Expertise: Moderate, including basic system navigation, experiment configuration, and data analysis tasks.

3. Participant

- Characteristics: Individuals who participate in stress-induction experiments conducted by researchers. Participants interact with the system only during experiment sessions when responding to stimuli presented by the system.
- Frequency of Use: Low to moderate
- Privilege Levels: Limited access to interact with experiment tasks during testing sessions. Participants cannot access system management features.
- Technical Expertise: Basic computer usage skills required to interact with experiment tasks and respond to stimuli.

2.3 Operating Environment

The Centralized Web-Based System for Stress Induction Methods operates within a secure and centralized environment. The following describes the hardware, software, and subsystems that support the system:

1. Hardware Platform
    - Server Requirements: The system will run on a dedicated server with a minimum of:
        - Processor: AMD Ryzen 5 or equivalent
        - RAM: 12 GB
        - Storage: 500 GB SSD
    - Client Devices:
        - Desktop computers or laptops for administers and researchers
        - Desktop computers or laptops for test sessions with participants

2. Operating System and Versions
    - Server OS: Windows Server
    - Client OS: Windows 10/11 for desktops and laptops

3. Operating System and Versions
    - Database Management System: MariaDB
    - Web Server: Apache HTTP Server
    - Application Framework: PHP
    - User Interface: Browser-based interface optimized for Chrome, Firefox, and Edge

2.4 Design and Implementation Constraints

The development of the Centralized Web-Based System for Stress Induction Methods must address several constraints that will limit the choices available to developers. These constraints are categorized below:

Type Of Constraints	**Constraints**
Hardware	Test sessions must be conducted using desktop computers or laptops to ensure accurate display of stimuli and reliable measurement of participant responses and reaction times.
Network	The system requires a stable internet connection to ensure experiment data can be recorded and stored without interruption. Network instability may affect the experiment session.
Software	The system must be developed as a web-based application using PHP, MariaDB database, and accessed through modern web browsers.
Security and Privacy	Participant data must be stored securely, and access must be restricted to authorized researchers and administrators only.
Language Requirements	The application must be developed using the English language to ensure consistency and usability for researchers

2.5 User Documentation

The Centralized Web-Based System for Stress Induction Methods will include comprehensive user documentation to support all user groups in effectively operating the system. The documentation components are as follows:

1. User Manuals
    - Administrator Manual: Includes instructions for managing user accounts, maintaining system settings, and monitoring system activities.
    - Researcher Manual: Provides detailed guidance on managing participant information, configuring experiment templates, conducting stress-induction sessions, and generating experiment reports.

2. Tutorials
    - Interactive Tutorials: Step-by-step walkthroughs that guide researchers on how to create participants, configure experiments, and run experiment sessions.
    - Video Tutorials: Visual demonstrations of key system features, such as starting experiment sessions, recording participant responses, and exporting experiment data.

3. Quick Reference Guides
    - One-page reference documents summarizing essential tasks for system users, such as creating experiment templates, running experiment sessions, and exporting research data.

2.6 Assumptions and Dependencies

The development and successful operation of the Centralized Web-Based System for Stress Induction Methods are based on the following assumptions and dependencies. Any changes to these factors may significantly affect the project's development and system performance.

Assumptions

1. Hardware Availability:

    - All end-user devices (desktop computers or laptops) meet the minimum hardware requirements such as at least 8GB RAM and a modern processor to ensure smooth system performance during experiment sessions.

2. Network Infrastructure:

    - The operating environment will have a stable and reliable internet connection to ensure that experiment data can be transmitted and stored in the database without interruption.

3. User Access and Training:

    - Researchers and administrators will receive sufficient guidance or training to operate the system effectively.
    - All system users will have unique login credentials and are responsible for maintaining the security of their accounts.

4. Software Compatibility:

    - The system will operate on supported operating systems such as Windows 10/11 or macOS.
    - Modern web browsers such as Brave, Google Chrome, Firefox, and Microsoft Edge will remain compatible with the system throughout its usage.

5. Organizational Support:

    - The research team and project supervisor will provide the necessary support and feedback to ensure the successful development and testing of the system.

6. Research Environment:

    - Experiment sessions will be conducted using desktop computers or laptops to ensure accurate presentation of stimuli and proper measurement of reaction time.
    - Participants will follow the instructions provided by the researcher during experiment sessions.

Dependencies

1. Database Management System (DBMS):

    - The system depends on the proper configuration and availability of the MariaDB database management system for storing participant data and experiment results.

2. Web Server and Hosting Environment:

    - The system requires a web server and hosting environment capable of supporting the web application and maintaining reliable access for researchers.

3. Development Tools:

    - The development of the system depends on tools such as Visual Studio Code for coding and Draw.io for system design diagrams.

4. Browser Compatibility:

    - The system depends on modern web browsers to function properly, including Brave, Google Chrome, Firefox, and Microsoft Edge.

5. Data Security and Privacy:

    - The system must follow appropriate data protection practices to ensure that participant information and experiment results are stored securely and accessed only by authorized users.

6. Stakeholder Feedback:

    - Continuous feedback from stakeholders, including researchers and the project supervisor, is necessary to ensure the system meets research requirements and project objectives.

3 System Features

3.1 USE CASE DIAGRAM



4. Other Non-functional Requirements

4.0 Performance Requirements

The system must provide a smooth and responsive interaction to ensure that the researchers can conduct the test, and the participants can perform the test efficiently. The system should respond to user actions such as button clicks, and page navigation.

Database operations including storing participants information, retrieving test question, and recording participants reaction must be complete within 2 seconds to avoid delays during the test. Reaction time recording and response must be processed accurately.

4.1 Safety and Security Requirements

The system must ensure the security and confidentiality of research data and participant information. All sensitive data, including participant records and experiment results, must be stored securely in the system database.

User authentication must be implemented to restrict system access only to authorized administrators and researchers. Passwords must follow strong security standards, including a combination of uppercase letters, lowercase letters, numbers, and special characters.

To prevent data loss, the system must implement automatic data saving during experiment sessions and ensure secure data transmission between the client and server. Access to participant data must be restricted according to user roles.

Additionally, system activities such as user logins, experiment sessions, and data modifications should be logged to maintain accountability and traceability.

4.2 Software Quality Attributes

4.2.1 Adaptability:

The system is designed to be adaptable so that additional stress-induction methods or test modules can be integrated in the future. The flexible system architecture allows researchers to create new experiment templates and configure parameters without major system changes.

4.2.2 Maintainability:

The system is developed with a modular structure that allows developers to easily maintain and update the software. Bug fixes, feature improvements, and security updates can be implemented efficiently without affecting other system components.

4.2.3 Portability:

The system is a web-based platform that can be accessed through modern web browsers which support major operating system such as Windows and macOS.

4.2.4 Reliability:

The system ensures reliable operation by securely storing experiment data and participant information. The database system is designed to maintain data integrity and reduce the risk of data loss during experiment sessions.

4.2.5 Usability:

The system uses a user-friendly interface designed for individuals with basic computer knowledge, ensuring ease of use. It is accessible for any devices with internet connection, providing convenience and flexibility for users to connect and use the system anywhere.

4.2.6 Performance Efficiency:

The system is optimized to provide efficient performance during experiment sessions. It ensures that stimuli presentation, response recording, and data processing occur quickly and accurately to support real-time experiment execution.